Networking Core Protocols

Task 1 -> Introduction

This room is the third room in a series of four rooms about computer networking:

- Networking Concepts
- Networking Essentials
- Networking Core Protocols (this room)
- Networking Secure Protocols

Room Prerequisites

To benefit from this room, we recommend that you know the following:

ISO OSI model and layers

TCP/IP model and layers

Ethernet, IP, and TCP protocols

In other words, starting this room after Networking Concepts is the recommended approach.

Learning Objectives

By the time you finish this room, you will have learned about the following protocols:

. WHOIS

. DNS

. HTTP and FTP

. SMTP, POP3, and IMAP

Click on the Start AttackBox button at the top; click on the Start Machine button belowt to start the attached virtual machine (VM).

▶ Start Machine

Give them about 2-3 minutes to boot up. Once the two machines are ready, we need to start the terminal on the AttackBox to follow along starting with Task 4.

Task 2 -> DNS: Remembering Addresses

Do you remember the IP addresses of your favourite websites? Unless it is a private IP address of a local device, no one needs to worry about memorizing IP addresses. This is in part due to the Domain Name System (DNS), which is responsible for properly mapping a domain name to an IP address.

DNS operates at the Application Layer, i.e., Layer 7 of the ISO OSI model. DNS traffic uses UDP port 53 by default and TCP port 53 as a default fallback. There are many types of DNS records; however, in this task, we will focus on the following four:

A record: The A (Address) record maps a hostname to one or more IPv4 addresses. For example, you can set example.com to resolve to 172.17.2.172.

AAAA Record: The AAAA record is similar to the A Record, but it is for IPv6. Remember that it is AAAA (quad-A), as AA and AAA would refer to a battery size; furthermore, AAA

refers to Authentication, Authorization, and Accounting; neither falls under DNS.

CNAME Record: The CNAME (Canonical Name) record maps a domain name to another domain name. For example, www.example.com can be mapped to example.com or even to example.org.

MX Record: The MX (Mail Exchange) record specifies the mail server responsible for handling emails for a domain.

In other words, when you type example.com in your browser, your browser tries to resolve this domain name by querying the DNS server for the A record. However, when you try to send an email to test@example.com, the mail server would query the DNS server to find the MX record.

If you want to look up the IP address of a domain from the command line, you can use a tool such as nslookup. Consider the example in the terminal below where we look up example.com.

```
Terminal

user@TryHackMe$ nslookup www.example.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   www.example.com
Address: 93.184.215.14
Name:   www.example.com
Address: 2606:2800:21f:cb07:6820:80da:af6b:8b2c
```

The query above led to four packets. In the terminal below, we can see that the first and third packets send DNS queries for the A and AAAA records, respectively. The second and fourth packets show the DNS query responses.

```
Terminal
user@TryHackMe$ tshark -r dns-query.pcapng -Nn
    1 0.000000000 192.168.66.89 → 192.168.66.1 DNS 86 Standard query 0x2e0f A www.example.com OPT
    2 0.059049584 192.168.66.1 → 192.168.66.89 DNS 102 Standard query response 0x2e0f A www.example.com A 93.184.215.14 OP
    3 0.059721705 192.168.66.89 → 192.168.66.1 DNS 86 Standard query 0x96e1 AAAA www.example.com OPT
    4 0.101568276 192.168.66.1 → 192.168.66.89 DNS 114 Standard query response 0x96e1 AAAA www.example.com AAAA 2606:2800:
```

Answer the questions below

1 -> Which DNS record type refers to IPv6?

Ans-> AAAA

2 -> Which DNS record type refers to the email server?

Ans-> MX

Task 3 -> WHOIS

In the previous task, we covered how a domain name is resolved into an IP address. However, for this to happen, someone needs to have the authority to set the A, AAAA, and MX records, among other DNS records for the domain. Whoever registers a domain name is granted this power. Therefore, if you register example.com, you can set any valid DNS records for example.com.

You can register any available domain name for one or more years. You need to pay the annual fee, and you are required to provide accurate contact information as the registrant. This information is part of the data available via WHOIS records and is available publicly. (Although written in uppercase, WHOIS is not an acronym; it is pronounced who is.) However, don't worry if you want to register a domain without revealing your contact information publicly; you can use one of the privacy services that hide all your information from the WHOIS records.

You can look up the WHOIS records of any registered domain name using one of the online services or via the command-line tool whois, available on Linux systems, among others. As expected, a WHOIS record provides information about the entity that registered a domain name, including name, phone number, email, and address. In the screenshot shown below, you can see when the record was first created and when it was last updated. Moreover, you can find the registrant's name, address, phone, and email.

Example WHOIS record

```
Domain Name: [redacted]
Registry Domain ID: [redacted]
Registrar WHOIS Server: whois.corporatedomains.com
Registrar URL: www.cscprotectsbrands.com
Updated Date: 2020-10-19T15:26:24Z
Creation Date: 1988-10-12T00:00:00.000-04:00
Registrar Registration Expiration Date: 2024-10-11T00:00:00.000-04:00
Registrar: CSC CORPORATE DOMAINS, INC.
Registrar IANA ID: 299
Registrar Abuse Contact Email: domainabuse@cscglobal.com
Registrar Abuse Contact Phone: +1.8887802723
Domain Status: clientTransferProhibited http://www.icann.org/epp#clientTransferProhibited
Domain Status: serverDeleteProhibited http://www.icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited http://www.icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited http://www.icann.org/epp#serverUpdateProhibited
Registry Registrant ID:
Registrant Name: [redacted]
Registrant Organi[redacted]
Registrant Street[redacted]
Registrant City: [redacted]
Registrant State/[redacted]
Registrant Postal[redacted]
Registrant Countr[redacted]
Registrant Phone: [redacted]
Registrant Phone [redacted]
Registrant Fax: +[redacted]
Registrant Fax Ex[redacted]
Registrant Email: [redacted]
Registry Admin ID:
Admin Name: [redacted]
Admin Organi[redacted]
Admin Street [redacted]
Admin City: [redacted]
Admin State/[redacted]
Admin Postal [redacted]
Admin Countr[redacted]
Admin Phone: [redacted]
Admin Phone [redacted]
Admin Fax: +[redacted]
Admin Fax Ex[redacted]
Admin Email: [redacted]
Registry Tec[redacted]
Tech Name: R[redacted]
Tech Organiz[redacted]
Tech Street: [redacted]
Tech City: M[redacted]
```

In the terminal output below, we have used the whois command to look up a domain whose WHOIS record is protected by privacy protection.

```
Terminal

user@TryHackMe$ whois [REDACTED].com
[...]
Domain Name: [REDACTED].COM
Registry Domain ID: [REDACTED]
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2017-07-05T16:02:43Z
Creation Date: 1993-04-02T00:00:00Z
Registrar Registration Expiration Date: 2026-10-20T14:56:17Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
[...]
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
[...]
```

Answer the questions below

1 -> When was the x.com record created? Provide the answer in YYYY-MM-DD format.

Ans-> 1993-04-02

2 -> When was the twitter.com record created? Provide the answer in YYYY-MM-DD format.

Ans-> 2000-01-21

Task 4 -> HTTP(S):Accessing the Web

When you fire up your browser, you mainly use HTTP and HTTPS protocols. HTTP stands for Hypertext Transfer Protocol; the S in HTTPS stands for Secure. This protocol

relies on TCP and defines how your web browser communicates with the web servers.

Some of the commands or methods that your web browser commonly issues to the web server are:

GET retrieves data from a server, such as an HTML file or an image.

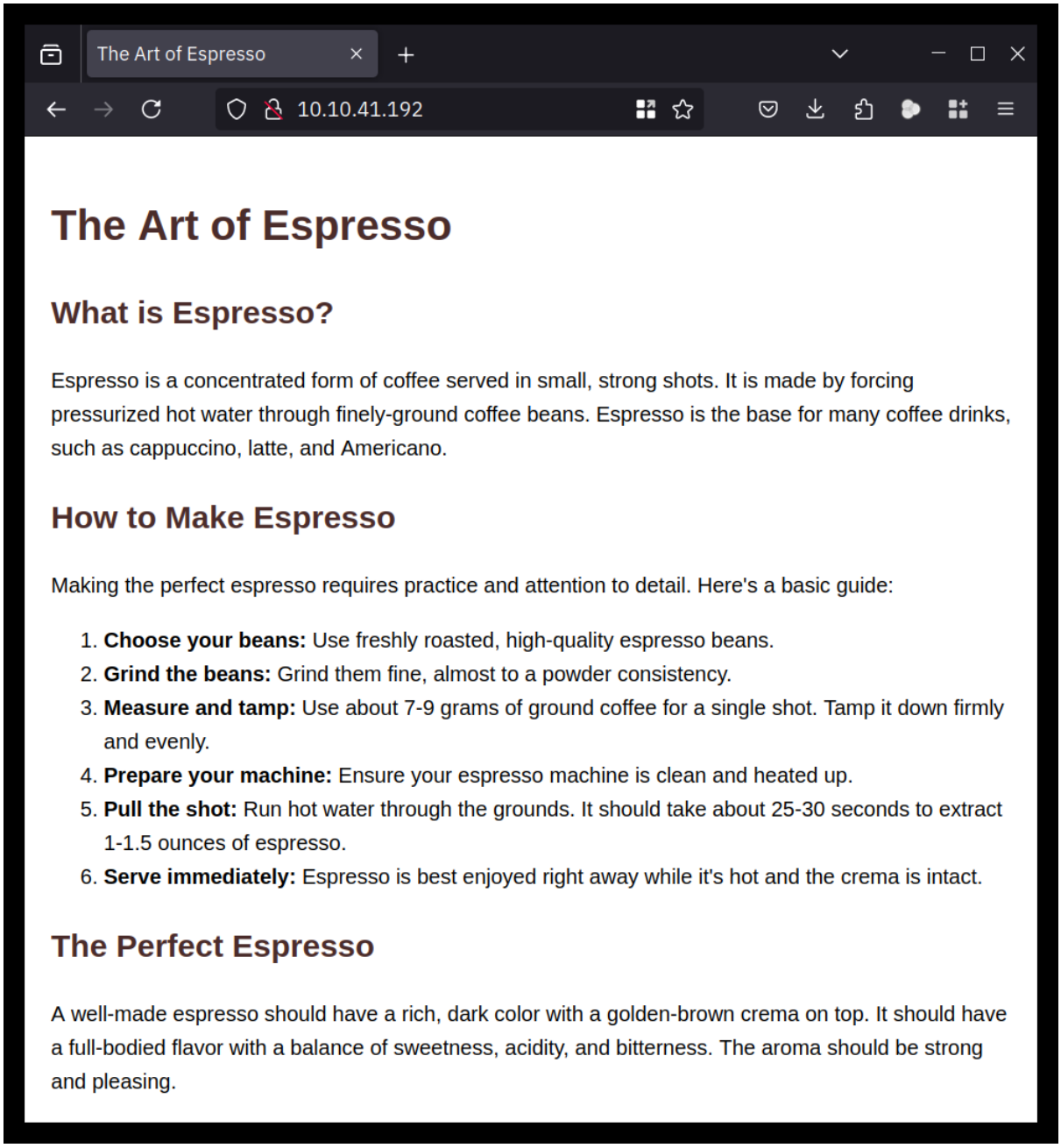
POST allows us to submit new data to the server, such as submitting a form or uploading a file.

PUT is used to create a new resource on the server and to update and overwrite existing information.

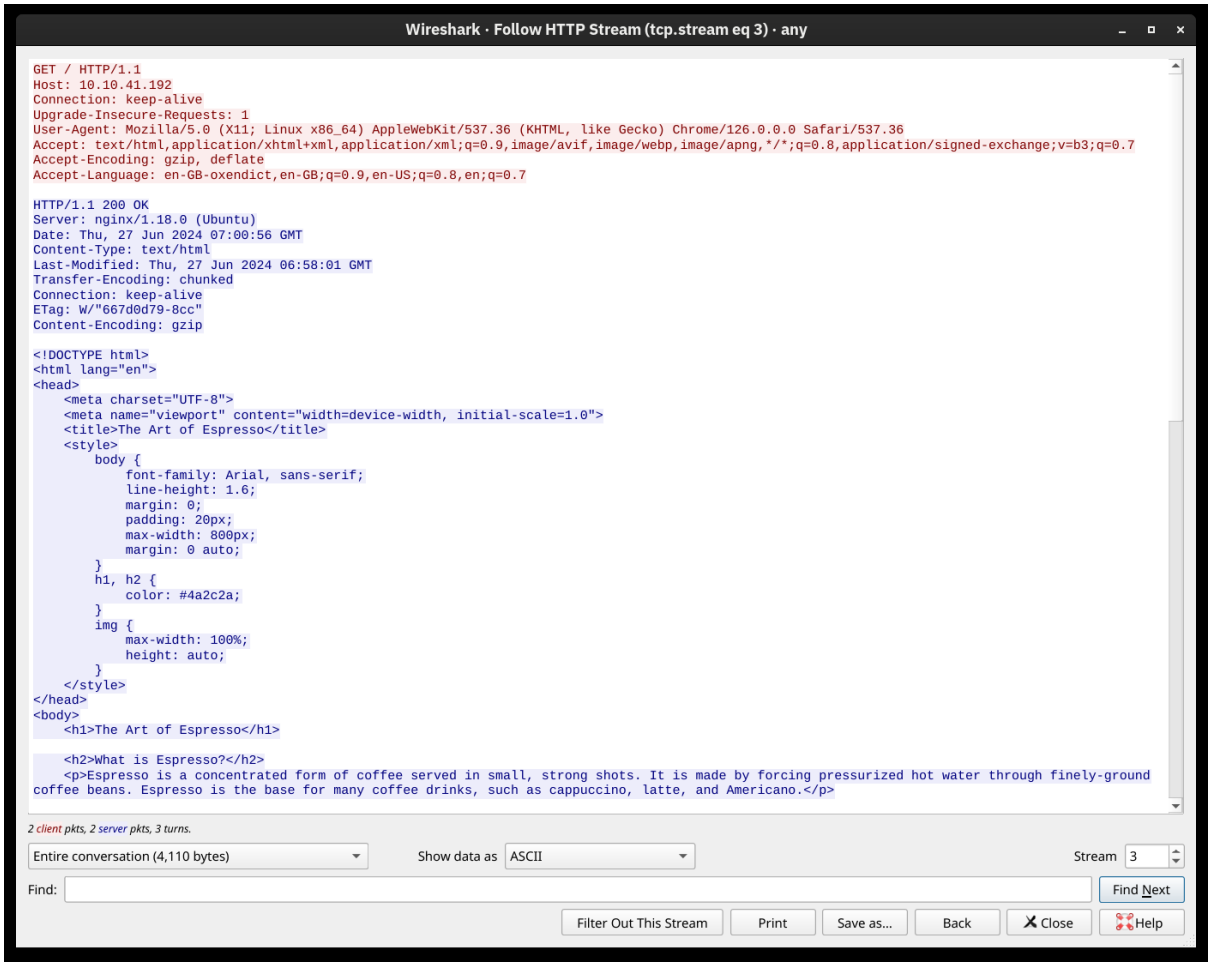
DELETE, as the name suggests, is used to delete a specified file or resource on the server.

HTTP and HTTPS commonly use TCP ports 80 and 443, respectively, and less commonly other ports such as 8080 and 8443.

In the following example, we use our Firefox browser to access the web server on MACHINE_IP. Our browser fetches the web page and displays it perfectly; however, we are interested in what happens behind the scenes.



Using Wireshark, we can examine the exchange between the Firefox browser and the web server more closely. The screenshot below from Wireshark shows the text sent by our browser in red and the web server response in blue. As you can tell, a lot of information is exchanged between the client and the server that does not get rendered to the user. Examples include the web server version and when the page was last modified.



As you remember from Networking Concepts, we used the telnet client to connect to the web server running on MACHINE_IP at port 80. We had to send a couple of lines: GET / HTTP/1.1 and Host: anything to get the page we wanted. (On some servers, you might get the file without sending Host: anything.) You can use this method to access any page and not just the default page /. To get file.html, you would send GET /file.html HTTP/1.1, for instance (GET /file.html might work depending on the web server in use). This approach is efficient for troubleshooting as you would be “talking HTTP” with the server.

Answer the questions below

1 -> Use telnet to access the file flag.html on MACHINE_IP. What is the hidden flag?

Ans-> THM{TELNET-HTTP}

Task 5 -> FTP:Transferring Files

Unlike HTTP, which is designed to retrieve web pages, File Transfer Protocol (FTP) is designed to transfer files. As a result, FTP is very efficient for file transfer, and when all conditions are equal, it can achieve higher speeds than HTTP.

Example commands defined by the FTP protocol are:

USER is used to input the username

PASS is used to enter the password

RETR (retrieve) is used to download a file from the FTP server to the client.

STOR (store) is used to upload a file from the client to the FTP server.

FTP server listens on TCP port 21 by default; data transfer is conducted via another connection from the client to the server.

In the terminal below we executed the command ftp MACHINE_IP to connect to the remote FTP server using the local ftp client. Then we went through the following steps:

We used the username anonymous to log in

We didn't need to provide any password

Issuing ls returned a list of files available for download

type ascii switched to ASCII mode as this is a text file

get coffee.txt allowed us to retrieve the file we want

The command exchange via the FTP client is shown in the terminal below.

```
Terminal
user@TryHackMe$ ftp MACHINE_IP
Connected to MACHINE_IP (MACHINE_IP).
220 (vsFTPd 3.0.5)
Name (MACHINE_IP:strategos): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (10,10,41,192,134,10).
150 Here comes the directory listing.
-rw-r--r--    1 0        0            1480 Jun 27 08:03 coffee.txt
-rw-r--r--    1 0        0              14 Jun 27 08:04 flag.txt
-rw-r--r--    1 0        0            1595 Jun 27 08:05 tea.txt
226 Directory send OK.
ftp> type ascii
200 Switching to ASCII mode.
ftp> get coffee.txt
local: coffee.txt remote: coffee.txt
```

```
local: coffee.txt remote: coffee.txt
227 Entering Passive Mode (10,10,41,192,57,100).
150 Opening BINARY mode data connection for coffee.txt (1480 bytes).
WARNING! 47 bare linefeeds received in ASCII mode
File may not have transferred correctly.
226 Transfer complete.
1480 bytes received in 8e-05 secs (18500.00 Kbytes/sec)
ftp> quit
221 Goodbye.
```

We used Wireshark to examine the exchanged messages more closely. The client's messages are in red, while the server's responses are in blue. Notice how various commands differ between the client and the server. For example, when you type ls on the client, the client sends LIST to the server. One last thing to note is that the directory listing and the file we downloaded are sent over a separate connection each.

Answer the questions below

1 -> Using the FTP client ftp on the AttackBox, access the FTP server at MACHINE_IP and retrieve flag.txt. What is the flag found?

Ans-> THM{FAST-FTP}

Task 6 -> SMTP: Sending Email

As with browsing the web and downloading files, sending email needs its own protocol. Simple Mail Transfer Protocol (SMTP) defines how a mail client talks with a mail server and how a mail server talks with another.

The analogy for the SMTP protocol is when you go to the local post office to send a package. You greet the employee, tell them where you want to send your package, and provide the sender's information before handing them the package. Depending on the country you are in, you might be asked to show your identity card. This process is not very different from an SMTP session.

Let's present some of the commands used by your mail client when it transfers an email to an SMTP server:

HELO or EHLO initiates an SMTP session

MAIL FROM specifies the sender's email address

RCPT TO specifies the recipient's email address

DATA indicates that the client will begin sending the content of the email message

. is sent on a line by itself to indicate the end of the email message

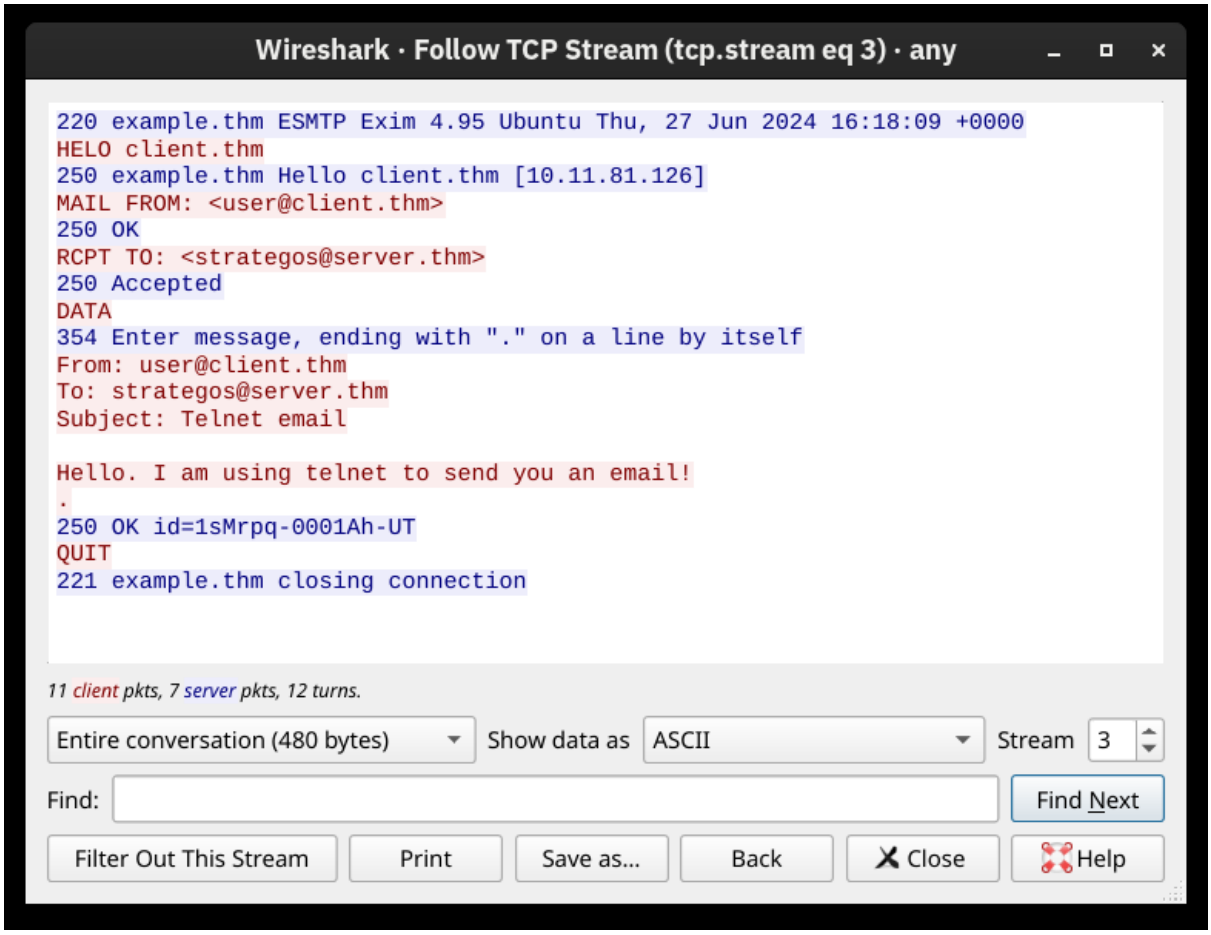
The terminal below shows an example of an email sent via telnet. The SMTP server listens on TCP port 25 by default.

```
Terminal
user@TryHackMe$ telnet MACHINE_IP 25
Trying MACHINE_IP...
Connected to MACHINE_IP.
Escape character is '^]'.
220 example.thm ESMTP Exim 4.95 Ubuntu Thu, 27 Jun 2024 16:18:09 +0000
HELO client.thm
250 example.thm Hello client.thm [10.11.81.126]
MAIL FROM: <user@client.thm>
250 OK
RCPT TO: <strategos@server.thm>
250 Accepted
DATA
354 Enter message, ending with "." on a line by itself
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email

Hello. I am using telnet to send you an email!
.
250 OK id=1sMrpq-0001Ah-UT
QUIT
221 example.thm closing connection
```

Obviously, sending an email using telnet is quite cumbersome; however, it helps you better understand the commands that your email client issues under the hood. The Wireshark capture shows the exchange in colours; the client's messages are in red, while the server's responses are in blue.

The data exchanged between the client and the SMTP server as captured by Wireshark.



Now that we have covered some basic HTTP, FTP, and SMTP commands, you should have gained a solid understanding of how protocols are designed and used. It should be effortless to learn how other text-based protocols, such as POP3 and IMAP, work.

Answer the questions below

1 -> Which SMTP command indicates that the client will start the contents of the email message?

Ans-> DATA

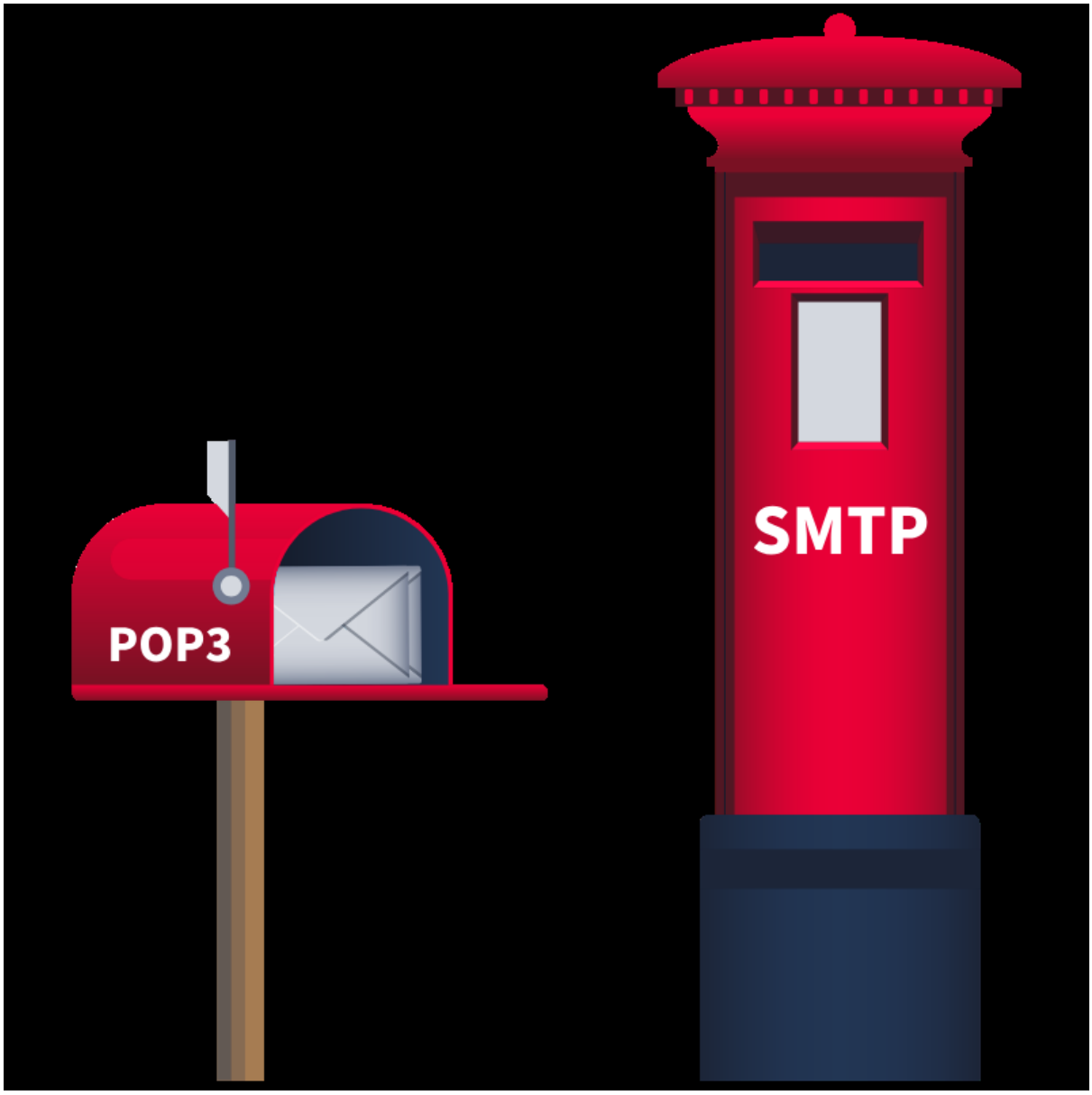
2 -> What does the email client send to indicate that the email message has been fully entered?

Ans-> (.)Dott

Task 7 -> POP3: Receiving Email

You’ve received an email and want to download it to your local mail client. The Post Office Protocol version 3 (POP3) is designed to allow the client to communicate with a mail server and retrieve email messages.

Without going into in-depth technical details, an email client sends its messages by relying on SMTP and retrieves them using POP3. SMTP is similar to handing your envelope or package to the post office, and POP3 is similar to checking your local mailbox for new letters or packages.



Some common POP3 commands are:

USER <username> identifies the user

PASS <password> provides the user's password

STAT requests the number of messages and total size

LIST lists all messages and their sizes

RETR <message_number> retrieves the specified message

DELE <message_number> marks a message for deletion

QUIT ends the POP3 session applying changes, such as deletions

In the terminal below, we can see a POP3 session over telnet. Since the POP3 server listens on TCP port 110 by default, the command to connect to the TELNET port is telnet MACHINE_IP 110. The exchange below retrieves the email message sent in the previous task.

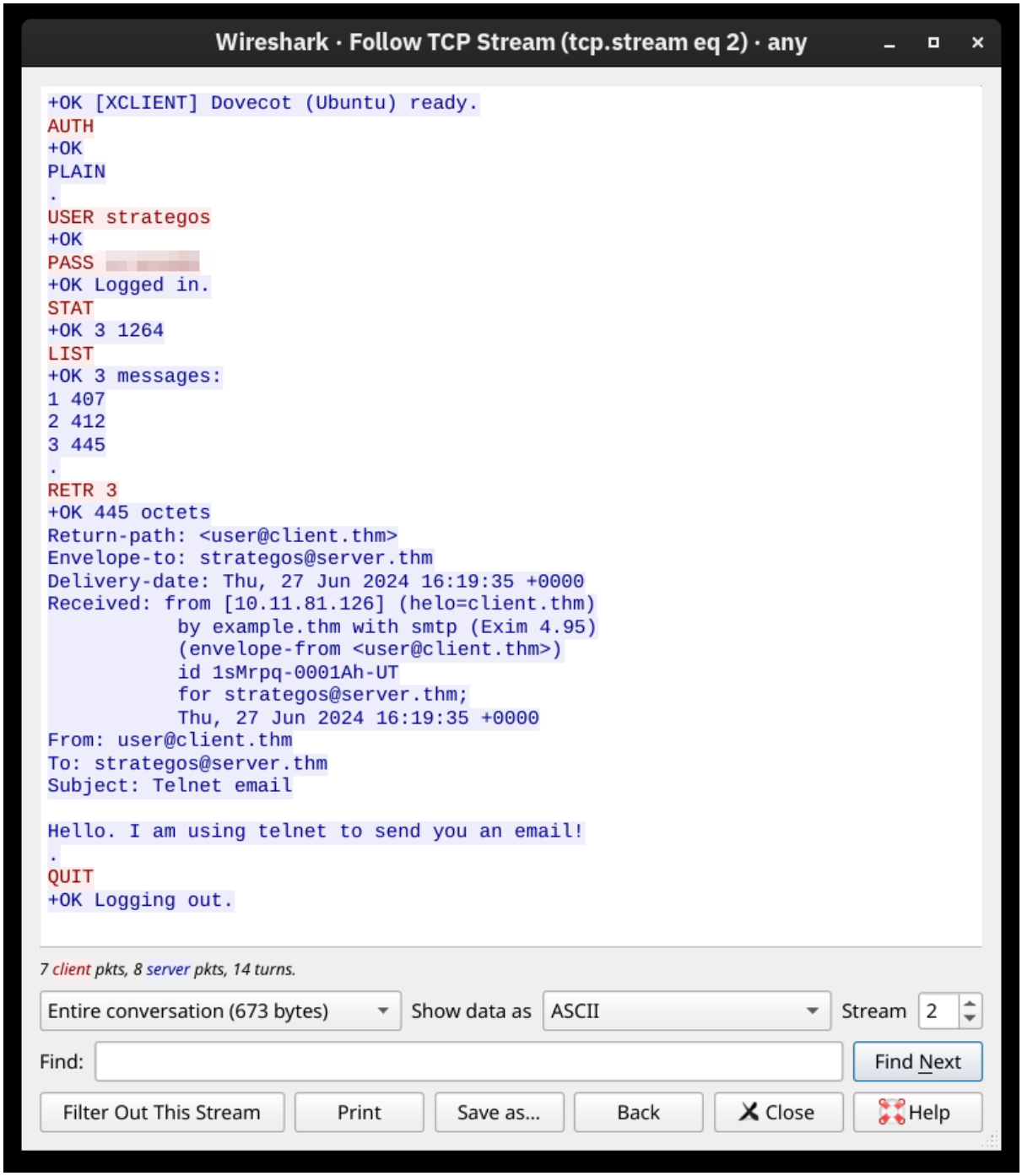
```
Terminal
user@TryHackMe$ telnet MACHINE_IP 110
Trying MACHINE_IP...
Connected to MACHINE_IP.
Escape character is '^]'.
+OK [XCLIENT] Dovecot (Ubuntu) ready.
AUTH
+OK
PLAIN
.
USER strategos
+OK
PASS
+OK Logged in.
STAT
+OK 3 1264
LIST
+OK 3 messages:
1 407
2 412
3 445
.
```

```
RETR 3
+OK 445 octets
Return-path: <user@client.thm>
Envelope-to: strategos@server.thm
Delivery-date: Thu, 27 Jun 2024 16:19:35 +0000
Received: from [10.11.81.126] (helo=client.thm)
        by example.thm with smtp (Exim 4.95)
        (envelope-from <user@client.thm>)
        id 1sMrpq-0001Ah-UT
        for strategos@server.thm;
        Thu, 27 Jun 2024 16:19:35 +0000
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email

Hello. I am using telnet to send you an email!
.
QUIT
+OK Logging out.
Connection closed by foreign host.
```

Someone capturing the network packets would be able to intercept the exchanged traffic. As per previous Wireshark captures, the commands in red are sent by the client, and the lines in blue are the server's. It is also clear that someone capturing the traffic can read the passwords.

The data exchanged between the client and the POP3 server as captured by Wireshark.



Connecting to a POP3 server requires authentication. Use the following login credentials when needed:

Username: linda

Password: Pa$$123

Answer the questions below

1 -> Looking at the traffic exchange, what is the name of the POP3 server running on the remote server?

Ans-> Dovecot

2 -> Use telnet to connect to MACHINE_IP's POP3 server. What is the flag contained in the fourth message?

Ans-> THM{TELNET_RETR_EMAIL}

Task 8 -> IMAP: Synchronizing Email

POP3 is enough when working from one device, e.g., your favourite email client on your desktop computer. However, what if you want to check your email from your office desktop computer and from your laptop or smartphone? In this scenario, you need a protocol that allows synchronization of messages instead of deleting a message after retrieving it. One solution to maintaining a synchronized mailbox across multiple devices is Internet Message Access Protocol (IMAP).

IMAP allows synchronizing read, moved, and deleted messages. IMAP is quite convenient when you check your email via multiple clients. Unlike POP3, which tends to

minimize server storage as email is downloaded and deleted from the remote server, IMAP tends to use more storage as email is kept on the server and synchronized across the email clients.

The IMAP protocol commands are more complicated than the POP3 protocol commands. We list a few examples below:

LOGIN <username> <password> authenticates the user

SELECT <mailbox> selects the mailbox folder to work with

FETCH <mail_number> <data_item_name> Example fetch 3 body[] to fetch message number 3, header and body.

MOVE <sequence_set> <mailbox> moves the specified messages to another mailbox

COPY <sequence_set> <data_item_name> copies the specified messages to another mailbox

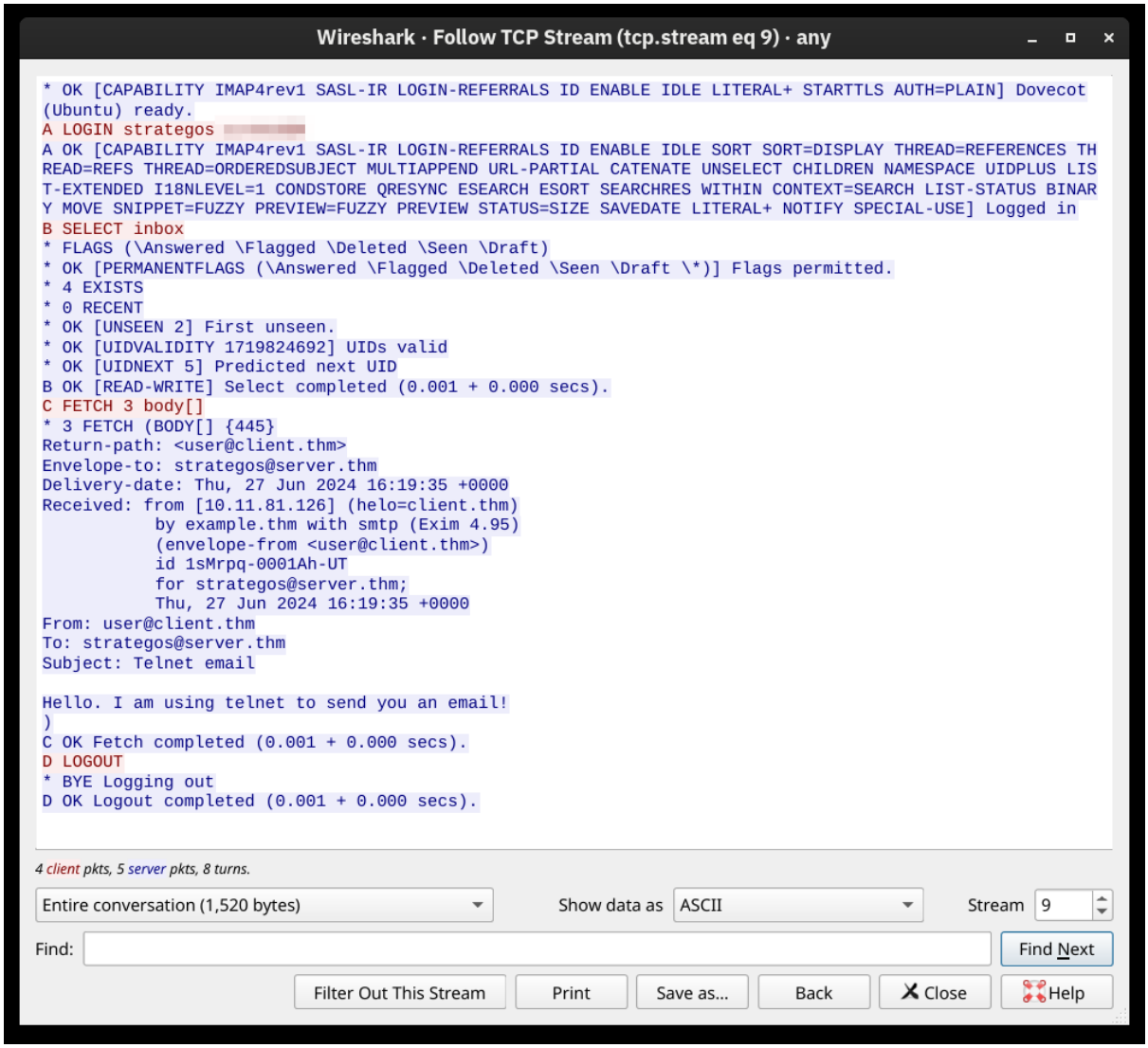
LOGOUT logs out

Knowing that the IMAP server listens on TCP port 143 by default, we will use telnet to connect to MACHINE_IP's port 143 and fetch the message we sent in an earlier task.

```
Terminal
user@TryHackMe$ telnet 10.10.41.192 143
Trying 10.10.41.192...
Connected to 10.10.41.192.
Escape character is '^]'.
* OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE LITERAL+ STARTTLS AUTH=PLAIN] Dovecot (Ubuntu) ready.
A LOGIN strategos
A OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SORT SORT=DISPLAY THREAD=REFERENCES THREAD=REFS THREAD=
B SELECT inbox
* FLAGS (\Answered \Flagged \Deleted \Seen \Draft)
* OK [PERMANENTFLAGS (\Answered \Flagged \Deleted \Seen \Draft \*)] Flags permitted.
* 4 EXISTS
* 0 RECENT
* OK [UNSEEN 2] First unseen.
* OK [UIDVALIDITY 1719824692] UIDs valid
* OK [UIDNEXT 5] Predicted next UID
B OK [READ-WRITE] Select completed (0.001 + 0.000 secs).
C FETCH 3 body[]
* 3 FETCH (BODY[] {445}
Return-path: <user@client.thm>
Envelope-to: strategos@server.thm
Delivery-date: Thu, 27 Jun 2024 16:19:35 +0000
Received: from [10.11.81.126] (helo=client.thm)
        by example.thm with smtp (Exim 4.95)
        (envelope-from <user@client.thm>)
        id 1sMrpq-0001Ah-UT
        for strategos@server.thm;
        Thu, 27 Jun 2024 16:19:35 +0000
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email

Hello. I am using telnet to send you an email!
)
C OK Fetch completed (0.001 + 0.000 secs).
D LOGOUT
* BYE Logging out
D OK Logout completed (0.001 + 0.000 secs).
Connection closed by foreign host.
```

The screenshot below shows the exchanged messages between the client and the server as seen from Wireshark. The client only needed to send four commands, shown in red, and the “long” server responses are shown in blue.



Answer the questions below

1 -> What IMAP command retrieves the fourth email message?

Ans-> FETCH 4 Body[]

Task 9 -> Conclusion

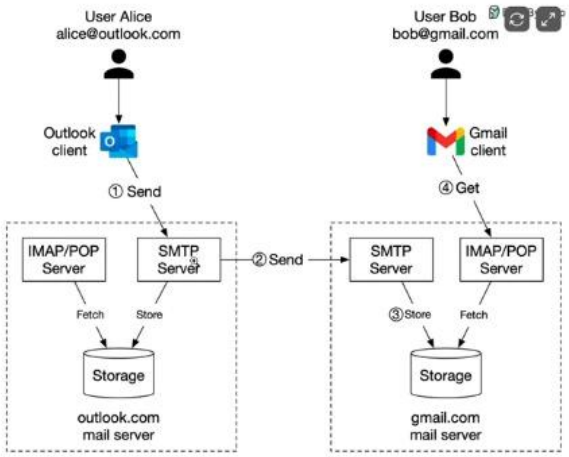
In the previous room, we discussed the TELNET protocol; this room focused on other fundamental protocols: DNS, HTTP,

FTP, SMTP, POP3, and IMAP. With the protocols covered, we now have a better understanding of how domain names are resolved, how web pages are served, and how email is sent and received. Another primary purpose of this room is to give you a good understanding of how a protocol functions behind the graphical interfaces.

The table below summarizes the default port numbers of the protocols we have covered so far.

Protocol	Transport Protocol	Default Port Number
TELNET	TCP	23
DNS	UDP or TCP	53
HTTP	TCP	80
HTTPS	TCP	443
FTP	TCP	21
SMTP	TCP	25
POP3	TCP	110
IMAP	TCP	143

what happens when Alice sends an email to Bob.



1. Alice logs in to her Outlook client, composes an email, and presses "send". The email is sent to the Outlook mail server. The communication protocol between the Outlook client and mail server is SMTP.

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Networking Core Protocols completed!

7060115632 completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
9	88	7

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